

TASK 2: FRONTEND SPA WITH REACT

Fetch API & Routing Implementation

Objective

The objective of this task is to develop a Single Page Application (SPA) using React.js that communicates with the backend REST API developed in Task 1. The application provides a user-friendly interface for managing student records and performs CRUD (Create, Read, Update, Delete) operations through API calls.

Technologies Used

- **Frontend Framework:** React.js
- **Routing:** React Router DOM
- **Languages:** JavaScript (ES6), HTML5, CSS3
- **Backend Integration:** Node.js Backend API, MongoDB Database
- **HTTP Client:** Fetch API / Axios

Project Description

A Student Management System was developed using React.js. The application allows users to:

- View all student records
- Add new students
- Update existing student details
- Delete student records
- Navigate between pages using React Router

The frontend communicates with the backend REST API running on port 5000 and displays data dynamically.

Features Implemented

1. Student List Page

- Displays all students retrieved from the backend API.
- Data is fetched using API requests.
- Shows Name, Email, and Department information.

2. Add Student Page

- Provides a form to add a new student.
- Sends POST requests to the backend.
- Updates the student list after successful submission.

3. Edit Student Page

- Allows modification of existing student details.
- Sends PUT requests to update data in MongoDB.

4. Delete Student Functionality

- Removes student records from the database.
- Sends DELETE requests to the backend API.

5. Routing

The following routes were implemented:

Route	Description
/	Home Page
/add	Add Student Page
/edit/:id	Edit Student Page

API Endpoints Used

GET /students
Get All Students

POST /students
Create Student

PUT /students/:id
Update Student

DELETE /students/:id
Delete Student

Project Structure

```
student-management-frontend
├── public/
├── src/
│   ├── pages/
│   │   ├── Home.js
│   │   ├── AddStudent.js
│   │   └── EditStudent.js
│   ├── App.js
│   └── index.js
├── package.json
└── README.md
```

Screenshots

The following application views represent the functionality and execution of the frontend system:

- 1. Home Page
- 2. Add Student Form
- 3. Student Added Successfully
- 4. Edit Student Page

- 5. Updated Student Record
- 6. Delete Student Functionality
- 7. Backend Running
- 8. Frontend Running
- 9. React Routing

(Insert Screenshots Here)

Output

The application successfully performs CRUD operations and communicates with the backend API. Student information is stored in MongoDB and displayed dynamically on the frontend interface.

Conclusion

The React-based Student Management System was successfully developed using React Router and API integration. The application demonstrates client-server communication, routing, state management using React Hooks, and CRUD operations. This task provided practical experience in building a frontend SPA that interacts with a backend REST API.